



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2010

YEAR 12

ASSESSMENT TASK #3

Mathematics

General Instructions

- Reading Time – 5 Minutes
- Working time – 120 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Answer in simplest exact form unless otherwise stated.
- Full marks may not be awarded for careless or badly arranged work.

Total Marks – 82

- Attempt questions 1 – 3
- All questions are **NOT** of equal value.
- Each question is to be returned in a separate bundle.

Examiner: *A. Fuller*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Total marks 82

Attempt questions 1 to 3

Answer each **Question** in a **Separate** writing booklet

(Use a SEPARATE writing booklet)

Question 1 (26 marks)

(a) Express $\frac{5\pi}{6}$ in degrees. 1

(b) Find the following correct to 2 decimal places: 2

(i) $\log_e \frac{3}{2}$

(ii) $\sin 2^c$

(c) Simplify $e^{3 \ln x}$ 1

(d) Differentiate the following with respect to x : 5

(i) $1 - 2x^2$

(ii) $2 \sin x^2$

(iii) e^{1-2x}

(iv) $\frac{\cos 2x}{x}$

(v) $(1 - 2 \ln x)^2$

(e) State a primitive (indefinite integral) of:

5

(i) x^{100}

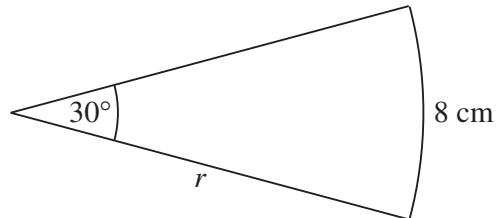
(ii) e^{100x}

(iii) $\sqrt{100x}$

(iv) $\frac{100 + x^2}{x^2}$

(f) Find the radius of a sector which has an arc length of 8 cm that subtends an angle of 30° at the centre.

2



(g) (i) Find all the values for x for which $4 \cos x + 2 = 0$ where $0 \leq x \leq 2\pi$.

4

(ii) Hence sketch the graph $y = 4 \cos x + 2$ for $0 \leq x \leq 2\pi$ marking clearly where it intersects with the x and y axes.

(h) A function is defined by $f(x) = x^3 - 3x^2 - 12$.

(i) Find the coordinates of the stationary points of the graph $y = f(x)$,
and determine their nature.

(ii) Hence sketch the graph of $y = f(x)$.

(iii) From the graph, or otherwise, for what values of x is $y = f(x)$ increasing?

(iv) From the graph, or otherwise, how many real solutions does $x^3 - 3x^2 - 12 = 0$
have?

(Use a SEPARATE writing booklet)

Question 2 (28 marks)

(a) Find the exact value of the following:

4

(i) $\int_0^{\frac{\pi}{2}} \sin \frac{x}{2} dx$

(ii) $\int_2^6 \frac{dx}{2+x}$

(b) At what point on the curve $y = \ln 2x$ is the gradient of the tangent $\frac{1}{2}$?

2

(c) For a certain continuous function $f(x)$, $f(2) = 2$ and $f'(2) = -1$.

2

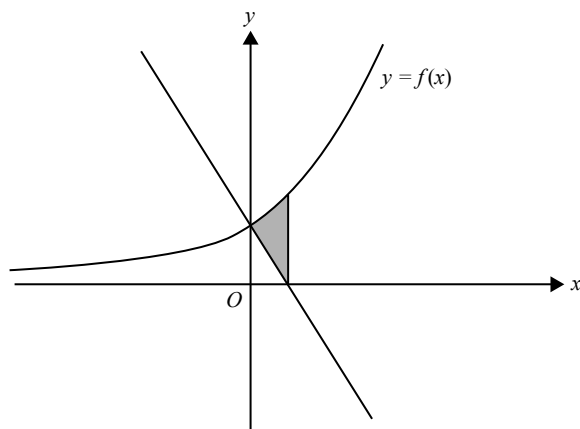
If $g(x) = x.f(x)$, evaluate $g'(2)$.

(d) Find the volume of the solid of revolution when the area bound by the curve $y = x^2 + 1$, the x -axis, the y -axis and the line $x = 3$ is rotated about the x -axis.

3

- (e) The graph of $y = f(x)$ where $f(x) = e^{\frac{x}{2}} + 1$ is shown below. The normal to the graph of $y = f(x)$ where it crosses the y -axis is also shown.

5



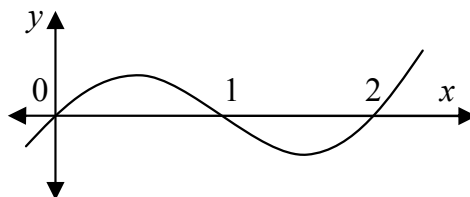
- (i) Find the equation of the normal to the graph of $y = f(x)$ where it crosses the y axis.
- (ii) Find the exact area of the shaded region.

- (f) If $\int_1^3 (2f(x) + 5)dx = 8$ determine the exact value of $\int_1^3 f(x)dx$.

2

(g) The graph of $y = x(x - 1)(x - 2)$ is given below.

6



(i) Expand and simplify $x(x - 1)(x - 2)$

(ii) Show that $y = x(x - 1)(x - 2)$ has an inflexion point when $x = 1$.

(iii) Show that $\int_0^2 x(x - 1)(x - 2)dx = 0$

(iv) $\int_0^1 x(x - 1)(x - 2)dx = \frac{1}{4}$.

Without evaluating the integral what is the value of $\int_1^2 x(x - 1)(x - 2)dx$?

(h) A particle moves in a straight line in such a way that its displacement in metres from the origin after t seconds is given by $x = 2t^3 + 3t^2 - 36t + 10$.

4

(i) In which direction is the particle moving initially?

(ii) When does the particle come to rest?

(iii) What is the displacement of the particle after 3 seconds?

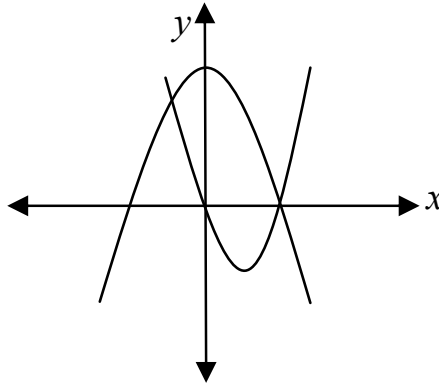
(iv) What distance has the particle travelled in the first 3 seconds?

(Use a SEPARATE writing booklet)

Question 3 (28 marks)

- (a) The functions $y = 4 - x^2$ and $y = x^2 - 2x$ are sketched below on the same axes.

6



- (i) Copy the above sketch into your answer booklet and label where each function meets the x and y axes.
- (ii) Find the points of intersection of the two functions.
- (iii) Shade on your diagram in part (i) the region which satisfies the following inequalities: $y \geq x^2 - 2x$, $y \leq 4 - x^2$, $y \geq 0$
- (iv) Calculate the area of the shaded region.

(b) (i) Show that $\sec^2 x + \tan^2 x = 2 \sec^2 x - 1$

4

(ii) By writing $\sec x$ as $(\cos x)^{-1}$ show that $\frac{d(\sec x)}{dx} = \sec x \tan x$

(iii) Hence, or otherwise, find $\int (\sec x + \tan x)^2 dx$

(c) (i) Copy and complete the table in your answer booklet

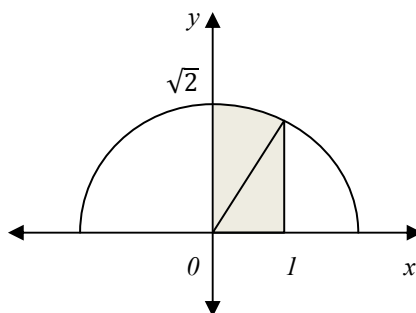
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x	0	$\frac{1}{2}$	1
$\sqrt{2-x^2}$			

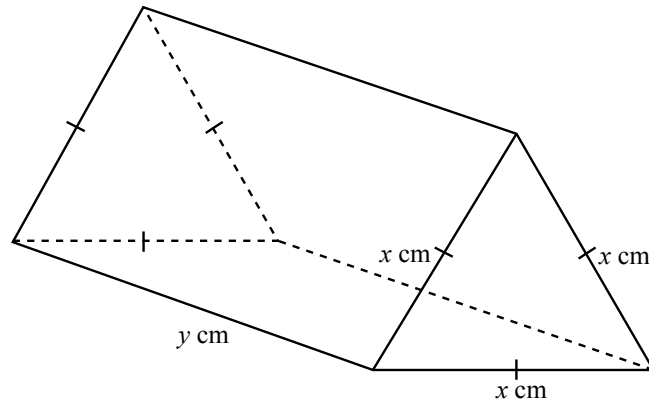
(ii) Use Simpson's Rule with three function values to approximate

$\int_0^1 \sqrt{2-x^2} dx$ to 2 decimal places.

(iii) By considering the area below, find the exact value of $\int_0^1 \sqrt{2-x^2} dx$



- (d) A plastic brick is made in the shape of a right triangular prism. The triangular end is an equilateral triangle with side length x cm and the length of the brick is y cm.



The volume of the brick is 1000 cm^3 .

- (i) Show that the area of the equilateral triangle is given by $\frac{\sqrt{3}x^2}{4}$.
- (ii) Find an expression for y in terms of x .
- (iii) Show that the total surface area, $A \text{ cm}^2$, of the brick is given by
- $$A = \frac{4000\sqrt{3}}{x} + \frac{\sqrt{3}x^2}{2}.$$
- (iv) Find the value of x for which the brick has minimum total surface area.

(e) $A(5, 20)$, $B(30, 15)$, $C(20, -10)$ and D are the vertices of a quadrilateral $ABCD$. 7

Given that the diagonals AC and BD are perpendicular.

(i) Prove that the point D lies on the line $y = \frac{x}{2}$.

(ii) If also $AB = AD$, prove that the coordinates of D are $(-6, -3)$.

(iii) Prove that AC bisects BD .

(iv) What type of quadrilateral is $ABCD$?

End of paper

Mathematics

Assessment 3, 2010

Question 1:

a. $\frac{5\pi}{6} \times \frac{180}{\pi} = 150^\circ$ (1)

b.i. $\log_e(3/2) = 0.4054651081$
 $= 0.41$ (2dp) (1)

ii. $\sin 2^\circ = 0.9092974268$
 $= 0.91$ (2dp) (1)

c. $e^{3\ln x} = (e^{\ln x})^3$
 $= x^3$ (1)

di. $\frac{d}{dx} (1 - 2x^2) = -4x$ (1)

ii. $\frac{d}{dx} (2\sin x^2) = 2\cos x^2 \times 2x$
 $= 4x\cos x^2$ (1)

iii. $\frac{d}{dx} (e^{1-2x}) = -2(e^{1-2x})$
 $= -2e^{1-2x}$ (1)

iv. $\frac{d}{dx} \left(\frac{\cos 2x}{x} \right) = u$
 $-v$

$\frac{d}{dx} = \frac{vu' - uv'}{v^2}$

$u = \cos 2x$ $v = x$

$u' = -2\sin 2x$ $v' = 1$

$= \frac{-2x\sin 2x - \cos 2x}{x^2}$ (1)

v. $\frac{d}{dx} (1 - 2\ln x)^2$

$= 2(1 - 2\ln x) \times -\frac{2}{x}$

$= -\frac{4}{x} (1 - 2\ln x)$ (1)

e.i. $\int x^{100} dx$

$= \frac{x^{101}}{101} + c$ (1)

-1 for first instance of
no +c only

ii. $\int e^{100x} dx = \frac{1}{100} e^{100x} + c$ (1)

iii. $\int \sqrt{100x} dx = \int \sqrt{100} \cdot \sqrt{x} dx$

$= \int 10x^{1/2} dx$

$= \frac{10x^{3/2}}{3/2} + c$

$= \frac{20x^{3/2}}{3} + c$ (1)

iv. $\int \frac{100 + x^2}{x^2} dx = \int \frac{100}{x^2} + \frac{x^2}{x^2} dx$

$= \int (100x^{-2} + 1) dx$

$= \frac{100x^{-1}}{-1} + x + c$

$= x - \frac{100}{x} + c$ (2)

$$f. l = r\theta$$

$$\theta = 30^\circ = \frac{\pi}{6}$$

$$f(0) = 0 - 0 - 12 = -12$$

$$f(2) = 2^3 - 3(2)^2 - 12 = -16$$

$\therefore$ stat points are

$$(0, -12) \text{ and } (2, -16) \quad (1)$$

$$\theta = r \times \frac{\pi}{6} \quad (2)$$

$$\therefore r = \frac{48}{\pi}$$

$$(0, -12)$$

x	-1	0	1
$f'(x)$	7	0	-3

/ - \ $\therefore$ max.

$$g.i. 4\cos x + 2 = 0$$

$$0 \leq x \leq 2\pi$$

$$4\cos x = -2$$

$$\cos x = -1/2$$

$$\therefore x = 2\pi/3, 4\pi/3$$

S A
X C

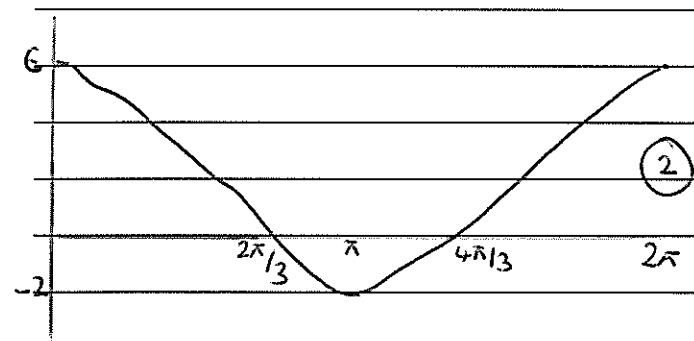
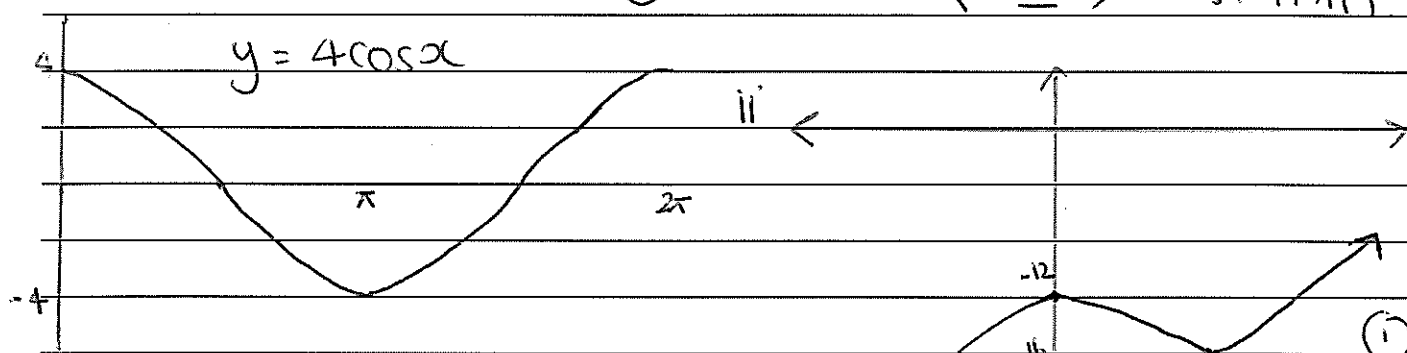
(2)

$$(2, -16)$$

x	1	2	3
$f'(x)$	-3	0	9

(1)

\ - / $\therefore$ min



iii. $f(x)$ is increasing for $x < 0, x > 2$ (2)

iv. 1 solution (1)

$$h.i. f(x) = x^3 - 3x^2 - 12$$

$$f'(x) = 3x^2 - 6x$$

for stat. points $f'(x) = 0$

$$\therefore 3x^2 - 6x = 0$$

$$3x(x - 2) = 0$$

$$\therefore x = 0, 2$$

$$\begin{aligned}
 \text{Q2 (a)(i)} \int_0^{\frac{\pi}{2}} \sin \frac{x}{2} dx &= \left[-2 \cos \frac{x}{2} \right]_0^{\frac{\pi}{2}} \\
 &= -2 \left[\cos \frac{\pi}{4} - \cos 0 \right] \\
 &= -2 \left[\frac{1}{\sqrt{2}} - 1 \right] \\
 &= 2 - \sqrt{2}.
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \int_2^6 \frac{dx}{2+x} &= \left[\ln(x+2) \right]_2^6 \\
 &= \ln 8 - \ln 4 \\
 &= \ln \frac{8}{4} \\
 &= \ln 2
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad y &= \ln 2x \\
 y' &= 2 \cdot \frac{1}{2x} = \frac{1}{x}
 \end{aligned}$$

$$\text{When } y' = \frac{1}{2}, \quad \frac{1}{x} = \frac{1}{2} \longrightarrow x = 2$$

$$\therefore \text{Point is } (2, \ln 4)$$

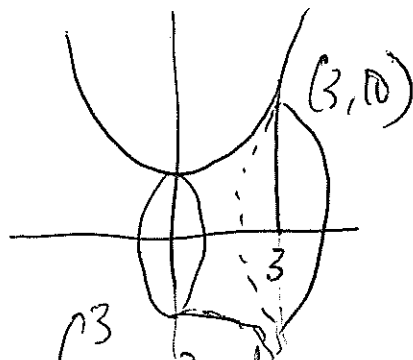
$$\begin{aligned}
 \text{(c)} \quad f(2) &= 2 \quad f'(2) = -1 \\
 g(x) &= x * f(x)
 \end{aligned}$$

$$\therefore \text{By Product Rule } g'(x) = f(x) \times 1 + x \times f'(x)$$

$$\begin{aligned}
 \therefore g'(2) &= f(2) + 2 f'(2) \\
 &= 2 + 2 \times -1 \\
 &= 0.
 \end{aligned}$$

$$(d) \quad y = x^2 + 1$$

$$\text{When } x = 3, y = 10$$



$$\begin{aligned}
 \therefore \text{Required volume} &= \pi \int_0^3 y^2 dx \\
 &= \pi \int_0^3 [x^2 + 1]^2 dx \\
 &= \pi \int_0^3 (x^4 + 2x^2 + 1) dx \\
 &= \pi \left[\frac{x^5}{5} + \frac{2}{3}x^3 + x \right]_0^3 \\
 &= \pi \left[\left[\frac{243}{5} + \frac{2}{3} \cdot 27 + 3 \right] - [0] \right] \\
 &= \pi \left\{ 48\frac{3}{5} + 18 + 3 \right\} \\
 &= 69\frac{3}{5} \pi \text{ cubic units}
 \end{aligned}$$

$$(e) \quad (i) \quad y = e^{\frac{x}{2}} + 1$$

$$y' = \frac{1}{2} e^{\frac{x}{2}}$$

$$\text{At } x=0, \quad y' = \frac{1}{2} e^0 = \frac{1}{2}$$

$$\therefore m_{\text{TANGENT}} = \frac{1}{2} \rightarrow m_{\text{NORMAL}} = -\frac{2}{1} = -2$$

$$\text{At } x=0, \quad y = e^0 + 1 = 2$$

$$\therefore \text{Equation of normal is } y = -2x + 2$$

i.e. $2x + y - 2 = 0$.

$$\begin{aligned} (ii) \text{ Required area} &= \int_0^1 (e^{\frac{x}{2}} + 1) dx - \int_0^1 (-2x + 2) dx \\ &= \int_0^1 (e^{\frac{x}{2}} + 1 + 2x - 2) dx \\ &= \int_0^1 (e^{\frac{x}{2}} + 2x - 1) dx \\ &= \left[2e^{\frac{x}{2}} + x^2 - x \right]_0^1 \\ &= [2e^{\frac{1}{2}} + 1 - 1] - [2e^0 + 0 - 0] \\ &= (2e^{\frac{1}{2}} - 2) \text{ sq. units} \end{aligned}$$

$$(1) \int_1^3 [2f(x) + 5] dx = 8$$

$$\therefore \int_1^3 2f(x) dx + \int_1^3 5 dx = 8$$

$$\text{ie } 2 \int_1^3 f(x) dx + [5x]_1^3 = 8$$

$$2 \int_1^3 f(x) dx + [15 - 5] = 8$$

$$\text{ie } 2 \int_1^3 f(x) dx = 8 - 10 \\ = -2$$

$$\text{ie } \int_1^3 f(x) dx = -1.$$

$$(9) \quad y = x(x-1)(x-2)$$

$$(i) \quad y = x(x^2 - 3x + 2) \\ = x^3 - 3x^2 + 2x$$

$$(ii) \quad y' = 3x^2 - 6x + 2 \\ y'' = 6x - 6$$

$$y'' = 0 \rightarrow 6x - 6 = 0 \rightarrow x = 1.$$

$\therefore$ Possible point of inflexion at $x=1$.

Check for change of sign of $f''(x)$ at $x=1$.

x	0	1	2
	-	0	+

ie there is a change of sign

$\therefore f(x)$ has point of inflexion at $x=1$

$$(iii) \quad \int_0^2 (x^3 - 3x^2 + 2x) dx = \left[\frac{x^4}{4} - x^3 + x^2 \right]_0^2 \\ = \left[\frac{16}{4} - 8 + 4 \right] - [0] \\ = 4 - 8 + 4 \\ = 0$$

$$(iv) \quad \text{Now } \int_0^2 f(x) dx = \int_0^1 f(x) dx + \int_1^2 f(x) dx$$

But $\int_0^2 f(x) dx = 0$ [from iii] and $\int_0^1 f(x) dx = \frac{1}{4}$ [given]

$\therefore \int_1^2 f(x) dx$ must be $-\frac{1}{4}$.

$$(h) \quad x = 2t^3 + 3t^2 - 36t + 10.$$

$$(i) \quad x' = 6t^2 + 6t - 36.$$

When $t=0$, $x' = -36$ (ie x' is negative)
 $\therefore$ Particle is moving to the left.

$$(ii) \quad x' = 0 \rightarrow 6t^2 + 6t - 36 = 0$$

$$\text{ie } t^2 + t - 6 = 0$$

$$\therefore (t+3)(t-2) = 0$$

$$\text{ie } t = -3, 2$$

But $t \geq 0$, $\therefore$ Particle comes to rest when $t=2$

$$(iii) \text{ When } t=3, \quad x = 54 + 27 - 108 + 10$$

$$= 91 - 108$$

$$= -17$$

$$(iv) \text{ When } t=0, \quad x = 10 \text{ metres}$$

$$\text{When it stops after 2 secs, } x = 16 + 12 - 72 + 10$$

$$= 38 - 72$$

$$= -34 \text{ metres}$$

} Particle has travelled 44 units in 1st 2 secs

$$\text{When } t=3, \quad x = -17 \text{ metres}$$

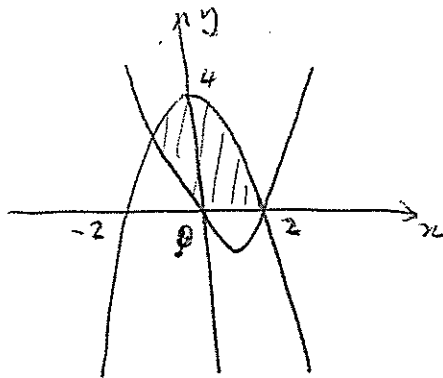
ie Particle has moved from -34 to -17 in the 3rd sec

$\therefore$ Particle has travelled 17 metres in 3rd second

$\therefore$ Distance travelled in 1st 3 secs $= 44 + 17 = 61 \text{ m}$

3(a)

(i)



(ii)

$$4 - x^2 = x^2 - 2x$$

$$2x^2 - 2x - 4 = 0$$

$$x^2 - x - 2 = 0$$

$$(x-2)(x+1) = 0$$

$$x = 2 \text{ or } x = -1$$

Point of intersection $(2, 0), (-1, 3)$

$$(iv) \text{ Area} = \int_{-1}^0 ((4-x^2) - (x^2-2x)) dx + \int_0^2 (4-x^2) dx$$

$$= \int_{-1}^0 (4+2x-2x^2) dx + \int_0^2 (4-x^2) dx$$

$$= \left[4x + x^2 - \frac{2}{3}x^3 \right]_{-1}^0 + \left[4x - \frac{1}{3}x^3 \right]_0^2$$

$$= [0] - \left[-4 + 1 - \frac{2}{3} \right] + \left[8 - \frac{8}{3} \right] - [0]$$

$$= 2\frac{1}{3} + 5\frac{1}{3}$$

$$= 7\frac{2}{3} \text{ units}^2$$

$$\begin{aligned} b) (i) \sec^2 x + \tan^2 x &= \sec^2 x + \sec^2 x - 1 \\ &= 2\sec^2 x - 1 \end{aligned}$$

$$(ii) \frac{d}{dx} (\sec x) = \frac{d}{dx} ((\cos x)^{-1})$$

$$= -1 (\cos x)^{-2} \cdot -\sin x$$

6

$$= \frac{\sin x}{\cos^2 x}$$

$$= \tan x \cdot \sec x$$

$$(iii) \int (\sec x + \tan x)^2 dx$$

$$= \int (\sec^2 x + 2 \sec x \tan x + \tan^2 x) dx$$

$$= \int (2 \sec^2 x + 2 \sec x \tan x - 1) dx$$

$$= 2 \tan x + 2 \sec x - x + c$$

(c) (i)

x	0	$\frac{1}{2}$	1
$\sqrt{2-x^2}$	$\sqrt{2}$	$\frac{\sqrt{3}}{2}$	1

$$(ii) \int_0^1 \sqrt{2-x^2} dx = \frac{1}{2} \left(\sqrt{2} + 4 \cdot \frac{\sqrt{3}}{2} + 1 \right)$$

$$= \frac{1}{2} (\sqrt{2} + 2\sqrt{3} + 1)$$

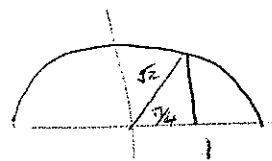
$$= \frac{1.284286}{2} = 0.642143$$

$$\approx 0.642143$$

$$(iii) \int_0^1 \sqrt{2-x^2} dx = \frac{1}{2} \cdot (\sqrt{2})^2 \cdot \frac{\pi}{4} + \frac{1}{2} \times 1 \times 1$$

$$= \frac{\pi}{4} + \frac{1}{2}$$

$$= 1.28539$$



5

$$(d) (i) \text{ Area of } \Delta = \frac{1}{2} x^2 \cdot \sin \frac{\pi}{3}$$

$$= \frac{\sqrt{3} x^2}{4}$$

$$(ii) \text{ Volume} = 1000 = \frac{\sqrt{3} x^2}{4} \times y.$$

$$\therefore y = \frac{4000 \sqrt{3}}{\sqrt{3} x^2}$$

$$(iii) A = 3 \times \frac{4000}{\sqrt{3} x^2} \times x + 2 \times \frac{\sqrt{3} x^2}{4}$$

$$= \frac{4000 \sqrt{3}}{x} + \frac{\sqrt{3} x^2}{2}$$

$$(iv) \text{ For min } A \quad \frac{d(A)}{dx} = 0.$$

$$\frac{dA}{dx} = -4000 \sqrt{3} x^{-2} + \sqrt{3} x \therefore 4000 \sqrt{3} \times -1 x^{-2} + \sqrt{3} x = 0.$$

$$\therefore \sqrt{3} x^3 - 4000 \sqrt{3} = 0$$

$$\therefore x^3 = 4000$$

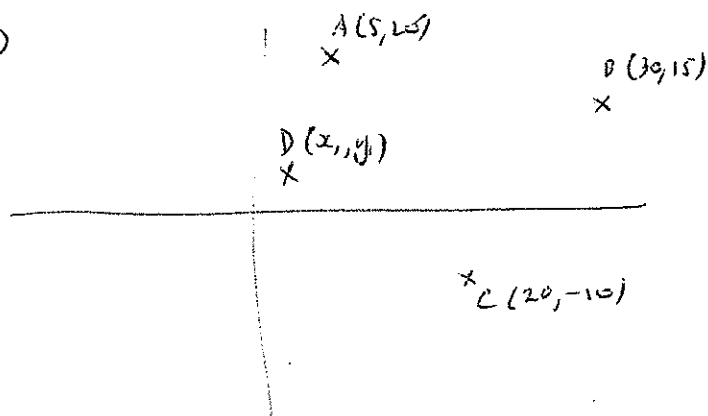
$$\therefore x = 10 \sqrt[3]{4}$$

$$\frac{d^2 A}{dx^2} = +8000 \sqrt{3} x^{-3} + \sqrt{3}.$$

$$\text{When } x = 10 \sqrt[3]{4}, \quad \frac{d^2 A}{dx^2} > 0$$

$$\therefore \text{Min surface area when } x = 10 \sqrt[3]{4}$$

(e)



3

6

$$(1) \quad m_{AC} = \frac{30}{-15} = -2$$

$$m_{BD} = \frac{15-y_1}{30-x_1}$$

$$m_{AC} \times m_{BD} = -1$$

$$\therefore -2 \times \frac{15-y_1}{30-x_1} = -1$$

$$\therefore \frac{15-y_1}{30-x_1} = \frac{1}{2}$$

$$\therefore 15-y_1 = 15 - \frac{1}{2}x_1$$

$$\therefore y_1 = \frac{1}{2}x_1$$

$$\therefore D \text{ lies on } y = \frac{1}{2}x$$

2

$$(ii) \quad AB = AD$$

$$\sqrt{(30-5)^2 + (15-20)^2} = \sqrt{(x_1-5)^2 + (y_1-20)^2}$$

$$\therefore \sqrt{625 + 25} = \sqrt{x_1^2 - 10x_1 + 25 + y_1^2 - 40y_1 + 400}$$

$$\therefore 650 = x_1^2 - 10x_1 + 25 + \frac{1}{4}x_1^2 - 20x_1 + 400$$

$$\therefore 225 = \frac{5}{4}x_1^2 - 30x_1$$

$$\therefore 900 = 5x_1^2 - 120x_1$$

$$\therefore 180 = x_1^2 - 24x_1$$

$$x_1^2 - 24x_1 - 180 = 0$$

$$(x_1 + 6)(x_1 - 30) = 0$$

$$\therefore x_1 = -6 \quad \text{or} \quad x_1 = 30$$

$$y_1 = -3 \quad y_1 = 15$$

$$\therefore D \text{ is } (-6, -3)$$

2

$$11 \quad -180$$

$$2 \quad -90$$

$$3 \quad -60$$

$$4 \quad -45$$

$$5 \quad -36$$

$$6 \quad -30$$

(ii) ~~M~~ $M_{BD} = (12, 6) = M$

$$m_{AM} = \frac{20-6}{5-12} = -2 = m_{AC}$$

2

$\therefore M$ lies on AC
 $\therefore AC$ bisects BD

(iv) $ABCD$ is a kite

1

